

Phase retrieval is strictly weaker than universal sorting once there are three rows

Candidate full implication-level answer to the open question in arXiv:2306.13111

11 August 2026

Status: candidate full result, likely valid. For every number of rows $n \geq 2$, an injective sorting encoder β_A forces the phase-retrieval map α_A to be injective, and the same implication holds with unchanged admissible bi-Lipschitz constants. The converse is an equivalence for $n = 2$, as proved in the source, but is strict for every $n \geq 3$ and $d \geq 2$. More sharply, every minimally redundant real phase-retrieval frame, with $D = 2d - 1$ measurements, fails to be a universal sorting key for every $n \geq 3$. Consequently a universal key for $n \geq 3$ requires $D \geq 2d$.

1 The source question

Let $A = [a_1 | \cdots | a_D] \in \mathbb{R}^{d \times D}$. The source compares

$$\alpha_A(x) = (|\langle x, a_j \rangle|)_{j=1}^D$$

on $\mathbb{R}^d / \{\pm 1\}$ with the columnwise sorting encoder

$$\beta_A(X) = \downarrow(XA), \quad X \in \mathbb{R}^{n \times d},$$

on $\mathbb{R}^{n \times d} / S_n$. The matrix A is called a *universal key* when β_A separates row-permutation orbits. Theorem 1 of the source proves that, for $n = 2$, A is a universal key if and only if it is a real phase-retrieval frame. The conclusion asks for the relationship between α_A and β_A when $n > 2$ [1, Conclusion].

The result below completely determines the implication relation between the two injectivity properties. It also gives a quantitative one-way relation and a new template-count obstruction.

2 Idea of the proof

There are two simple but complementary ideas.

First, insert a signal x into the n -row point cloud

$$X_x = (x, -x, 0, \dots, 0)^T.$$

Its permutation-orbit distance is exactly $\sqrt{2}$ times the sign-orbit distance of x , while sorting each projected column merely places $-|\langle x, a_j \rangle|$ and $|\langle x, a_j \rangle|$ around zeros. Thus phase retrieval sits as a scaled isometric slice of the sorting problem.

Second, let A be a full-spark frame with the minimal $2d - 1$ columns. Partition the columns into blocks of sizes $d - 1, d - 1, 1$. Normals to the first two blocks can be scaled so that their sum is

orthogonal to the last column. This produces three nonzero vectors r_1, r_2, r_3 with sum zero such that every measurement vector is orthogonal to at least one r_i . Along such a measurement, the three projections of $\{r_1, r_2, r_3\}$ have the form $\{0, t, -t\}$, so they are unchanged after negating the entire point cloud. The two clouds are nevertheless not row permutations. This is a three-point switching component tailored to every minimal phase-retrieval frame.

3 The phase-retrieval slice

Write

$$d_{\pm}(x, y) = \min\{\|x - y\|, \|x + y\|\}$$

and

$$d_{S_n}(X, Y) = \min_{P \in S_n} \|X - PY\|_F.$$

For $n \geq 2$, let $X_x \in \mathbb{R}^{n \times d}$ have first two rows $x, -x$ and all remaining rows zero.

Lemma 1. *For all $x, y \in \mathbb{R}^d$,*

$$d_{S_n}(X_x, X_y) = \sqrt{2} d_{\pm}(x, y)$$

and

$$\|\beta_A(X_x) - \beta_A(X_y)\|_F = \sqrt{2} \|\alpha_A(x) - \alpha_A(y)\|_2.$$

Proof. The squared matching cost between the row multisets $\{x, -x, 0, \dots, 0\}$ and $\{y, -y, 0, \dots, 0\}$ equals

$$2\|x\|^2 + 2\|y\|^2 - 2 \max_{P \in S_n} \sum_{i=1}^n \langle (X_x)_i, (PX_y)_i \rangle.$$

The maximum inner-product sum is $2|\langle x, y \rangle|$, achieved by matching the two nonzero antipodal pairs with the favorable sign. This gives

$$d_{S_n}(X_x, X_y)^2 = 2(\|x\|^2 + \|y\|^2 - 2|\langle x, y \rangle|) = 2d_{\pm}(x, y)^2.$$

For the j -th column, putting $s = \langle x, a_j \rangle$ gives

$$\downarrow(X_x a_j) = (-|s|, 0, \dots, 0, |s|)^T$$

when sorting is increasing; decreasing order only reverses the vector. The squared difference of the j -th sorted columns for x and y is twice the squared difference of the corresponding magnitudes. Summing over j proves the second identity. $\square$

Proposition 2. *Let $n \geq 2$. If β_A is injective on $\mathbb{R}^{n \times d}/S_n$, then α_A is injective on $\mathbb{R}^d/\{\pm 1\}$.*

More quantitatively, if

$$c d_{S_n}(X, Y) \leq \|\beta_A(X) - \beta_A(Y)\|_F \leq C d_{S_n}(X, Y)$$

for all X, Y , then

$$c d_{\pm}(x, y) \leq \|\alpha_A(x) - \alpha_A(y)\|_2 \leq C d_{\pm}(x, y)$$

for all x, y . Hence the optimal lower constant of α_A is at least that of β_A , and its optimal upper constant is at most that of β_A .

Proof. Restrict β_A to the point clouds X_x and apply Lemma 1. If $\alpha_A(x) = \alpha_A(y)$, then $\beta_A(X_x) = \beta_A(X_y)$; injectivity makes the row multisets equal, which forces $y = \pm x$. $\square$

4 Strictness for every $n \geq 3$

We use the standard real complement-property characterization: a real frame does phase retrieval if and only if every partition of its columns has a spanning side [2]. With exactly $2d - 1$ columns, this is equivalent to full spark. Indeed, full spark makes the side with at least d columns span; conversely, a dependent d -set and its $(d - 1)$ -element complement violate the complement property.

Theorem 3. *Let $d \geq 2$, $n \geq 3$, and $A = [a_1 | \dots | a_{2d-1}] \in \mathbb{R}^{d \times (2d-1)}$ be full spark. Then β_A is not injective on $\mathbb{R}^{n \times d} / S_n$.*

Consequently every minimally redundant real phase-retrieval frame fails to be a universal key as soon as $n \geq 3$.

Proof. Partition the column indices into disjoint sets

$$I \dot{\cup} J \dot{\cup} \{k\} = [2d - 1], \quad |I| = |J| = d - 1.$$

Full spark implies that the columns in each of I and J span a hyperplane. Choose nonzero normals

$$u \perp \text{span}\{a_i : i \in I\}, \quad v \perp \text{span}\{a_j : j \in J\}.$$

Both $\langle a_k, u \rangle$ and $\langle a_k, v \rangle$ are nonzero; otherwise a_k together with one of the $(d - 1)$ -blocks would be a dependent d -set. Define

$$r_1 = \langle a_k, v \rangle u, \quad r_2 = -\langle a_k, u \rangle v, \quad r_3 = -(r_1 + r_2).$$

Then

$$r_1 + r_2 + r_3 = 0, \quad a_i \perp r_1 \ (i \in I), \quad a_j \perp r_2 \ (j \in J), \quad a_k \perp r_3.$$

The vectors r_1, r_2 are nonzero. They are not collinear: otherwise the two block hyperplanes would coincide and would contain the $2d - 2$ columns indexed by $I \cup J$, contradicting full spark. Hence $r_3 \neq 0$ as well.

Let $X, Y \in \mathbb{R}^{n \times d}$ have row multisets

$$\{r_1, r_2, r_3, 0, \dots, 0\}, \quad \{-r_1, -r_2, -r_3, 0, \dots, 0\},$$

respectively. Fix a measurement column a_ℓ . It is orthogonal to at least one r_q . Since the three scalar projections sum to zero, their multiset is

$$\{\langle r_1, a_\ell \rangle, \langle r_2, a_\ell \rangle, \langle r_3, a_\ell \rangle\} = \{0, t, -t\}$$

for some t . Negating the three vectors leaves this multiset unchanged. The additional zero rows agree, so every sorted column agrees and $\beta_A(X) = \beta_A(Y)$.

Yet X and Y are not row permutations. If the three-element multiset $\{r_1, r_2, r_3\}$ equaled its negative, then an odd-cardinality sign-invariant multiset would have to contain zero: all nonzero elements pair as $r, -r$ with equal multiplicities. None of the r_i is zero, a contradiction. $\square$

Corollary 4. *Let $d \geq 2$ and $n \geq 3$. Every universal key $A \in \mathbb{R}^{d \times D}$ has*

$$D \geq 2d.$$

Proof. Proposition 2 makes A a phase-retrieval frame, so the real complement property gives $D \geq 2d - 1$. Equality would make A full spark and contradict Theorem 3. $\square$

Corollary 5. *At the level of injectivity, the relationship is:*

<i>regime</i>	<i>relationship</i>
$n = 2$	$\beta_A \text{ injective} \iff \alpha_A \text{ injective}$ (source theorem),
$d = 1$	$\beta_A \text{ injective} \iff \alpha_A \text{ injective}$ for every n ,
$n \geq 3, d \geq 2$	$\beta_A \text{ injective} \implies \alpha_A \text{ injective}$, and the converse is false.

Proof. Only the one-dimensional statement remains. A frame in $\mathbb{R}$ does phase retrieval exactly when some $a_j \neq 0$. Sorting the corresponding nonzero scalar multiple of the rows recovers their unordered multiset, so the same condition makes β_A injective for every n .

For $n \geq 3, d \geq 2$, full-spark frames with $2d - 1$ columns exist, for example Vandermonde frames. They phase retrieve but fail universality by Theorem 3. $\square$

5 Current bounds and a sharp small case

The 2026 follow-up by Dym, Wellershoff, Tsoukanis, Levy, and Balan proves that every full-spark A is universal when

$$D \geq n(d - 1) + 1$$

and gives a different logarithmic lower bound [3, Theorems 4 and 5]. Its numerical section reports failure to find separating keys in the cases

$$(n, d, D) = (3, 2, 3), (3, 3, 5), (3, 4, 7),$$

exactly the pattern $n = 3, D = 2d - 1$, but states this only as numerical evidence [3, Sec. IV]. Theorem 3 proves that failure for every $d \geq 2$, for every full-spark choice of the columns, and indeed for every $n \geq 3$.

Combining Corollary 4 with the follow-up’s full-spark upper bound gives the exact small case

$$D_{\min}(n = 3, d = 2) = 4.$$

More generally, the present result improves the universal lower bound to $D \geq 2d$ throughout the regime $n \geq 3, d \geq 2$; depending on n, d , the follow-up’s logarithmic obstruction may be stronger.

6 Proof audit, scope, and novelty

The argument uses no genericity and no numerical inference. The only external structural input is the standard real complement-property theorem; the equivalence between minimal phase retrieval and full spark is reproved above. The construction is homogeneous and works with repeated zero rows, so the passage from three rows to arbitrary $n \geq 3$ is exact.

This is a full answer to the source’s implication-level injectivity question, plus a quantitative one-way result. It does not solve the separate neural network implementation problem mentioned in the same conclusion, nor does it determine the optimal number of templates for every n, d .

Searches through 11 August 2026 covered the source title and exact concluding sentence, “phase retrieval”, “universal key”, “sorting-based permutation-invariant embedding”, and the later literature. The closest paper is arXiv:2510.22186v2, which proves broader upper and lower bounds and records the $n = 3, D = 2d - 1$ failures only numerically. The three-point switching idea is related to classical discrete tomography [4], especially in the planar case, but no source located states the frame-uniform theorem above, the $D \geq 2d$ corollary, or the exact scaled phase-retrieval slice. Novelty confidence is therefore moderate.

References

- [1] R. Balan and E. Tsoukanis, *Relationships between the Phase Retrieval Problem and Permutation Invariant Embeddings*, arXiv:2306.13111v2 (2023), <https://arxiv.org/abs/2306.13111>.
- [2] R. Balan, P. G. Casazza, and D. Edidin, *On signal reconstruction without phase*, Applied and Computational Harmonic Analysis 20 (2006), 345–356.
- [3] N. Dym, M. Wellershoff, E. Tsoukanis, D. Levy, and R. Balan, *Quantitative Bounds for Sorting-Based Permutation-Invariant Embeddings*, arXiv:2510.22186v2 (2026), <https://arxiv.org/abs/2510.22186>.
- [4] J. Matoušek, A. Přívětivý, and P. Škovroň, *How many points can be reconstructed from k projections?*, SIAM Journal on Discrete Mathematics 22 (2008), 1605–1623.